

Post-Quantum Cryptography Migration

What's New Since 2026-08-31

Compiled by the Spaced Research Ledger

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This report covers what changed in the post-quantum migration between the last full edition, dated 31 August 2026, and 3 September 2026. It is a supplement, not a replacement. The nineteen parts of the full collection remain the record, and this document reads against them.

The news falls into seven sections. Section 1 covers certificates, identity and the standards plumbing beneath them. Section 2 covers the network gear that carries the new handshakes, where the biggest change of the period sits. Section 3 covers cloud, storage and enterprise platforms. Section 4 covers libraries, silicon and testing tools.

Section 5 covers operational technology, devices and vehicles. Section 6 covers finance and payments. Section 7 covers government guidance.

Two developments matter most. Cisco shipped fully quantum-safe site-to-site tunnels by adding post-quantum signatures to IKEv2, closing the last classical link in its IPsec chain [1]. And DigiCert came down on the side of pure ML-DSA certificates over composite ones [2]. That position will shape what certificate authorities issue next.

A third thread runs through several sections. Standard-library work in Go, PostgreSQL and Docker means many systems now get hybrid key exchange without anyone asking for it [3] [4] [5]. Five items conflict with earlier findings. They are reported under Unresolved Conflicts in their sections rather than resolved by guesswork.

1. Certificates, Identity and Standards Plumbing

Certificates bind a public key to a name, and everything from web browsing to code signing depends on them. Post-quantum certificates are much larger than today's. The industry has been arguing about whether to ship them pure, or as composites that carry a classical signature too.

DigiCert has taken a position. Its migration guidance now recommends ML-DSA-only certificates as the default, rather than composite ML-DSA certificates [2]. It cites convergence across the industry and working implementations in OpenSSL 3.5, BoringSSL and Java 24 [2]. It also cites the CA/Browser Forum's own focus on ML-DSA [2].

A small provider has gone further on the timestamp side. BeatQuantum launched an RFC 3161 timestamp authority on 1 September [6]. It signs tokens with pure ML-DSA-87 or SLH-DSA, with no classical component at all, under its own root [6].

Microsoft's certificate services documentation, updated 25 August, extends its phased plan. ML-DSA remains available in phase one [7]. ML-KEM and composite algorithms remain planned for phase two [7]. The support

plan now reaches the enrollment web services, the device enrollment service and the revocation responder [7].

Kerberos is getting a cleanup. An individual IETF draft from June proposes retiring RSA key transport, the weakest Diffie-Hellman group and SHA-1 from PKINIT [8]. It also adds a checksum field that can change algorithms later [8].

The plumbing beneath all of this moved too. wolfPKCS11 release 2.1 added ML-DSA and ML-KEM support in June [9]. That replaces the earlier description of those features as build-time flags.

The OpenSC libp11 releases of July and August added a full provider path for ML-KEM, ML-DSA, SLH-DSA and Falcon keys [10]. That covers key generation, signing and decapsulation [10]. No PKCS#11 3.3 revision or errata for external-mu ML-DSA signing had appeared as of 3 September, so version 3.2 remains current [11].

2. Networks, Appliances and VPNs

Network equipment is where post-quantum key exchange first went mainstream, because one firewall can protect every connection behind it. The remaining gap has been authentication. The tunnels were quantum-safe for secrecy but still trusted classical signatures to prove who was on the other end.

Cisco closed that gap. IOS XE 26.2 adds native ML-DSA signatures for IKEv2 mutual authentication in all three parameter sets [1]. The ML-KEM hybrid key exchange arrived in 26.1. Together they make a site-to-site IPsec tunnel post-quantum end to end, on the 8000 Series Secure Routers among others [1]. Juniper's Junos 26.2R1 widened its own SSH protection to the ACX, EX, MX, QFX and SRX platforms [12].

The application delivery vendors moved on certificates and hardware. F5's partner program has validated twelve certificate lifecycle integrations from eight partners [13]. They automate renewals and can roll out hybrid post-quantum certificates on BIG-IP [13]. A10 Networks announced the Thunder 3360S on 2 September, a mid-range appliance described as ready for future post-quantum support [14]. It also presented at a Taiwanese enterprise post-quantum summit in August [15].

The secure access vendors are shipping too. Netskope's Next Gen SWG now supports the X25519MLKEM768 hybrid exchange in commercial use, both client to proxy and proxy to server [16]. It arrived through an OpenSSL 3.5 upgrade and sits behind a tenant-level switch [16]. Zscaler documents inline inspection of traffic that uses hybrid or pure post-quantum key encapsulation, with automatic detection of ML-KEM groups [17]. Akamai opened its post-quantum behaviors to Standard TLS hostnames in limited availability on 13 August [18].

Citrix's best-practice paper for the third quarter covers NetScaler 14.1 build 51.72 and later [19]. It says both default SSL profiles bind the X25519_MLKEM768 curve, with TLS 1.3 required for negotiation [19]. Looking ahead, the Internet Architecture Board is holding a workshop in Prague on 11 and 12 October on deploying post-quantum authentication [20]. The oversized certificate chains of EAP-TLS are named as a problem to solve [20].

Unresolved Conflicts

Two disagreements sit in this section. Netskope's earnings call describes post-quantum algorithms integrated across more than 120 data centers in the second quarter [21]. Its product announcement describes commercial support behind a feature switch [16]. Earlier documentation described the same capability as a beta in release 137. Product documentation for a specific release would settle which description is current.

Citrix's own product documentation says hybrid key exchange is available only for enhanced SSL profiles and must be bound by hand [22]. Its later best-practice paper says both default profiles bind the curve automatically [19]. The build number in the newer paper suggests the default changed between the two documents. A release note naming the change would settle it.

3. Cloud, Storage and Enterprise Platforms

Most enterprise software inherits its cryptography from a library or a language runtime. When those move, whole product lines move with them, often without a product announcement.

That inheritance is now visible. Go 1.27.1 ships ML-DSA in its standard library, in all three parameter sets [3]. That was the gating dependency the Sigstore project had named for post-quantum signing. Docker Engine and CLI releases in the 29 series are built with Go 1.26 toolchains [5]. So outbound TLS connections to registries negotiate hybrid key exchange by default, unless a build overrides the curve preferences [5].

PostgreSQL 18 renames its curve setting and adds X25519 to the default [4]. It can be configured for the X25519MLKEM768 group when linked against OpenSSL 3.5, something earlier versions could not easily do [4].

Storage and enterprise platforms followed. NetApp's ONTAP 9.17.1 supports post-quantum pre-shared keys for IPsec authentication [23]. ONTAP 9.18.1 supports ML-KEM, ML-DSA and SLH-DSA for SSL when FIPS mode is disabled, with negotiation in both directions [24]. SAP added hybrid key exchange for TLS connections to its HANA cloud service in July [25]. On the confidential computing side, Intel said in March that ML-DSA based attestation for TDX is in development, alongside ML-KEM key negotiation [26].

4. Libraries, Silicon and Testing Tools

The algorithms are only as safe as their implementations. This section covers the optimized libraries that make them fast, the formal verification that makes them trustworthy, and the tools that hunt for timing leaks.

The native implementations reached milestones. mldsa-native version 2.0.0, released 3 August, is the first stable release [27]. It carries AVX2 and Neon assembly backends, super-optimizer integration, and completed formal verification of both the assembly and the C [27]. mlkem-native now includes a native RISC-V vector backend alongside its Arm, x86 and POWER backends [28].

Timing analysis advanced on two fronts. A mentorship project extended liboqs's constant-time tooling with two memory checkers [29]. It found a real timing leak in NTRU Prime through a nonzero mask function [29]. A

separate toolchain automates four timing analysis tools across the NIST additional-signature candidates [30]. It maps where their findings overlap and where they differ [30].

Two smaller items round out the section. Cloudflare Radar offers a public widget that performs a TLS handshake against any website to test for hybrid key exchange support [31]. The amortized post-quantum combiner draft for the Messaging Layer Security protocol was updated on 21 August [32].

Unresolved Conflicts

The RISC-V backend in mlkem-native is production code with vector instructions [28]. The full collection recorded that all lattice acceleration work for RISC-V remained academic proposals rather than adopted extensions. Both can be true. A software backend using the standard vector extension is not a dedicated cryptographic extension. But the earlier statement no longer describes the whole picture.

5. Operational Technology, Devices and Vehicles

Industrial controllers, cameras, medical devices and cars run for a decade or more once installed. That makes them the hardest systems to migrate and the ones where a delay compounds.

The most sober assessment came from the power sector. An analysis of utility environments expects the IT-heavy portion to migrate by around 2030 [33]. That portion means internet-facing systems, PKI and VPNs. Gateways, communication infrastructure and field devices face a further three to four years beyond that, because of legacy protocols and real-time constraints [33].

A 2026 IEEE paper offers a migration framework for existing SCADA environments [34]. Its taxonomy covers coexistence, hybridization, dual-root trust and protected conduits [34]. Its sequence starts with gateways and management planes [34].

Federal buildings are in scope. The GSA's physical access control laboratory is expanding to test quantum-resistant technology for building access and employee identification cards [35]. In the camera world, a published architecture adds post-quantum mechanisms at session setup and authentication without altering existing networks or symmetric encryption [36]. A Canadian integrator advises buying camera and recorder hardware whose cryptography can be upgraded by firmware [37]. A buyer's guide for intercoms argues that door devices with ten-year lifespans should ship post-quantum firmware in 2026, citing the EU Cyber Resilience Act [38].

Healthcare guidance sharpened. A July review in *Frontiers in Health Services* recommends a staged, hybrid, crypto-agile migration for connected medical devices [39]. It calls for hybrid key establishment at gateways and cryptographic dependency information in premarket submissions, to match FDA expectations under the federal food and drug law [39]. The same review notes that FDA guidance does not yet require a formal cryptographic bill of materials [39].

Vehicles hit a hard limit. An August study of cellular vehicle-to-everything sidelink tested the NIST signatures against the transport block limits of the SAE and IEEE standards [40]. Falcon-512 was the only one that fit, at a mean latency near 52 milliseconds [40]. Even then it met the delivery threshold only at low traffic under line of sight [40]. Dilithium and SPHINCS+ exceeded capacity [40].

Payments hardware and satellites advanced. IDEMIA Secure Transactions topped Juniper Research's post-quantum competitor leaderboard for its accelerators and smart card implementations [41]. STMicroelectronics' ST54M secure mobile chip carries a hardware accelerator for ML-KEM and ML-DSA [42]. It completed Common Criteria and EMVCo certification testing, with full certification targeted for July [42].

On 2 September, Cyprus and Greece established a quantum-secure satellite data link using Sparkle's interconnect system and Hellas Sat's infrastructure [43]. A day earlier, Arqit, Es'hailSat and AIEE demonstrated quantum-safe encryption over live satellite networks from the Doha teleport [44]. They reported no performance penalty [44].

Galileo's authentication services fill in the background. The Signal Authentication Service began initial capability in December 2025 on two satellites [45]. Full constellation encryption is planned for the second half of 2026, and its interface documents are due this year [45]. Oscilloquartz has integrated Galileo's open navigation message authentication into three of its timing platforms [46].

Unresolved Conflicts

Verifiy's DOME product claims device-level zero-trust security with post-quantum algorithms for BACnet, Modbus, DNP3 and OPC UA, retrofitted onto legacy devices [47]. The full collection recorded that no vendor-shipped post-quantum integration for OPC UA had been found. A product page is a vendor claim. A customer deployment or an independent test would settle whether the OPC UA support is real and shipping.

6. Finance and Payments

Payment networks carry messages with strict size limits set decades ago. Whether a post-quantum signature fits inside those limits decides which algorithm the industry can use.

A closer reading changed the SWIFT picture. A simulation study of Australia's real-time payment infrastructure finds that SWIFT imposes no signature-size constraint on post-quantum signatures [48]. Authentication happens out of band at the messaging layer, so neither the signature nor the key travels in the message text [48].

That replaces the earlier recommendation of Falcon-512 as the only signature small enough for the message limit. The study now recommends ML-DSA-65 with ML-KEM-768 as the default pair across both the domestic platform and SWIFT routes [48]. Falcon-512 is reserved for the cases where in-band, size-constrained carriage is required [48].

The standards bodies published. X9 issued X9.150 for merchant-presented QR code payments on 14 July [49]. The codes are signed under the X9 financial PKI in a way it says prepares for post-quantum signatures [49]. EMVCo released the second of three position papers on quantum computing and chip cryptography in the first quarter [50].

PCI PTS HSM version 5.0 was published in May [51]. It adds post-quantum definitions and considerations [51]. It raises the minimum effective key strength for device-security keys to 128 bits and drops triple DES for device security [51].

7. Government Guidance

Governments set the deadlines the rest of this report is measured against. This period brought a delay in Europe and a pilot extension in the United Kingdom.

The EU's Quantum Act, originally expected in the second quarter of 2026, has slipped to a tentative fourth quarter [52]. The United Kingdom's NCSC has kept its post-quantum consultancy pilot open to qualified firms until 31 March 2027 [53]. It will then review the criteria and what it learned [53]. Expressions of interest have closed because of demand [53]. Its July workshop report names executive sponsorship, cost-of-delay messaging, early supplier engagement and cross-sector transparency as the themes that came out of the sessions [54].

About This Report

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